# MAST20026 - Real Analysis – Semester 1, 2025

| | | |
|---|---|---|
| **Subject Coordinator:** | TriThang Tran<br>trithang.tran@unimelb.edu.au<br>G44 Peter Hall Building | |
| **Subject Information:** | On the Canvas LMS – Check! | |
| **Lectures:** | Wednesday 11am-12pm | Lyle Theatre |
| | Thursday 3-4pm | Lyle Theatre |
| | Friday 3-4pm | Lyle Theatre |
| **Assessments:** | 5 written assignments (20% total)<br>Weekly quizzes (10%)<br>Final exam (70%) | |

# How to use these lecture slides

These slides are designed to facilitate the lecture presentation and note taking.

The full content of the course is not presented on these slides. Rather, plenty of blank space is provided for the lecturer to write out content and draw pictures during the lecture.

Students are strongly encouraged to attend/watch lectures regularly and take notes in the blank space provided.

Additionally, students should study and full digest the course material after each lecture. It may help to write out the main definitions and theorems from lecture and come up with relevant examples and counter-examples.

This subject has two main aims:

- To study how to write **mathematical proofs**;
- To study **Real Analysis**.

One important topic in Real Analysis is Fourier series.

A Fourier series takes the form

$$f(x) = a_0 + \sum_{k=1}^{\infty} (a_k \cos(kx) + b_k \sin(kx))$$

where $x$ is a variable and $a_k$ and $b_k$ are real number constants.

Fourier series have many important applications in mathematics and physics.
For instance, solutions to the heat equation in thermal dynamics can be represented as:

$$f(x,t) = \sum_{k=1}^{\infty} b_k \, \mathrm{e}^{\mathrm{kt}} \sin(kx)$$

Another example is the Weierstrass function

$$f(x) = \sum_{k=1}^{\infty} \frac{1}{2^k} \sin(3^k \pi x)$$

It is continuous at every real number $x$ but differentiable nowhere!

# Chapter 1:
# Proof and Mathematical Logic

## 1.1 Propositional Logic

Propositional logic is about variables and functions which take *two values*, called "True" and "False".

**Definition 1.1.** *A* <u>*statement*</u> *is a sentence or expression that is either true or false.*

A statement takes on the role of a logical variable (taking values **true** or **false**). We generally use lower case letters $p, q, r, \ldots$ to represent statements.

Some books use the word <u>proposition</u> statements rather than propositions.

**Example.** (Which of these are statements?)

- $1 + 1 = 3$
- $f(x)$ is continuous.
- 6 is the largest integer.
- Potato
- If it is Saturday, there is no MAST20026 lecture.

**Example.** (Which of these are statements?)

- $x > 2$

- For every $x \in \mathbb{Z}$, $x^2 \geq 0$.

- Every even number greater than 2 is the sum of two primes.

- $\begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$ is invertible and $1 + 1 = 3$.

**Connectives.** We can combine statements to give new statements, called compound statements. They are constructed from:

1. primitive statements (i.e. logical variables)
2. connectives
3. parentheses (to remove ambiguity)

Often, we use upper case letters $A, B, C, \ldots$ to denote compound statements.

The connectives we will consider are:

- negation
- implication
- conjunction
- biconditional
- disjunction

**Definition 1.2.** *Let $p$ be a statement. The* <u>*negation*</u> *of $p$ is the statement "not $p$" and is denoted by $\sim p$. The truth value of $\sim p$ is the opposite of $p$.*

**Examples**

Let $p$ be the statement "$5 > 0$".

Let $r$ be the statement "this lecture is boring".

**Definition 1.3.** *Let $p$ and $q$ be statements. The* <u>*conjunction*</u> *of $p$ and $q$ is the statement "$p$ and $q$" and is denoted by $p \wedge q$. The truth value of $p \wedge q$ is given by the following truth table.*

**Example.**

Let $p$ be the statement "$\begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$ is invertible." Let $q$ be the statement $1 + 1 = 3$.

In the English language, there are two different usages of the word "*or*".

The *exclusive* use of "or":

I will ride my bike or catch the train.

usually means that one or the other of these modes of transportation is used, but not both. This is **not** the mathematical usage of "or".

**Definition 1.4.** *Let $p$ and $q$ be statements. The* <u>*disjunction*</u> *of $p$ and $q$ is the statement "$p$ or $q$" and is denoted by $p \vee q$. The truth value of $p \vee q$ is given by the following truth table.*

Disjunction is an *inclusive* use of "or"; that is, it allows for the possibility that both statements are true.

**Example**

Let $p$ be the statement "$\begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$ is invertible." Let $q$ be the statement $1 + 1 = 3$.

**Example 1.5.** For which values of $p$ and $q$ are the following compound statements true:

(a) $(\sim p \wedge q) \wedge (p \vee q)$

(b) $(p \vee \sim p) \wedge (q \vee \sim q)$

**Definition 1.9.** *Let $p$ and $q$ be statements. The* <u>*implication*</u> *"$p$ implies $q$" and is denoted by $p \Rightarrow q$. The truth value of $p \Rightarrow q$ is given by the following truth table.*

**Example.** Let $p$ be the statement

"The temperature outside is colder than 0°C."

And let $q$ be the statement

"Brian wears a beanie when he is outside."

**Definition 1.10.** *Let $p$ and $q$ be statements. The* <u>*converse*</u> *of the implication $p \Rightarrow q$ is the implication $q \Rightarrow p$.*

In general, knowing $p \Rightarrow q$ tells us nothing about the truth value of the converse.

**Definition 1.11.** *Let $p$ and $q$ be statements. The* <u>*biconditional*</u> *"$p$ if and only $q$" is denoted by $p \Leftrightarrow q$. The truth value of $p \Leftrightarrow q$ is given by the following truth table.*

**Example.**

$$A \text{ is invertible if and only if } \det A \neq 0$$

**Definition 1.12.** *Let $p$ and $q$ be statements. The* <u>*contrapositive*</u> *of the implication $p \Rightarrow q$ is the implication $(\sim q) \Rightarrow (\sim p)$.*

**Example.**

If it is Saturday, then there is no MAST20026 lecture.

**Definition 1.7.** *Let $A$ be a compound statement. We say $A$ is a* <u>*tautology*</u> *when $A$ is always true regardless of the truth values of the simpler statements used to build $A$.*

**Definition 1.8.** *Let $A$ be a compound statement. We say $A$ is a* <u>*contradiction*</u> *when $A$ is always false regardless of the truth values of the simpler statements used to build $A$.*

**Definition 1.13.** *Let $r$ and $s$ be statements. We say $r$ and $s$ are* <u>*logically equivalent*</u> *when $r \Leftrightarrow s$ is a tautology. When $r$ and $s$ are logically equivalent we write $r \equiv s$.*

**Exercise 1.14.** Prove $\sim (p \vee q) \equiv (\sim p) \wedge (\sim q)$.

## 1.2 First-Order Logic

Consider the following:

- $f(x)$ is continuous and differentiable

- $x^2 \geq 9$

- The set $A$ has ten elements.

Mathematical sentences whose truth depend on variables (mathematical objects) are called <u>conditions</u>. The three examples above are conditions.

In all likelihood, you have seen conditions before, though perhaps not explicitly.

Conditions are hiding in the background when we use set-builder notation.

For example, consider the set

$$\{x | x^2 \geq 9\}$$

When we consider a condition for a particular value in a domain, the result is a statement. Thus we can combine conditions using the connectives. For instance:

$$\sim p(x) \qquad p(x) \vee q(x) \qquad p(x) \wedge q(x)$$

$$p(x) \Rightarrow q(x) \qquad p(x) \Leftrightarrow q(x)$$

**Example.**

$p(A)$ : "$A$ is invertible" $\qquad$ $q(A)$ : "$\det(A) \neq 0$"

Truth values partition the domain into those for which the corresponding statement is *true* and those for which the corresponding statement is *false*.

Let $q$ be a condition. There are four (overlapping) possibilities for this partition:

1. $q(x)$ is true for every $x$ in the domain.
2. $q(x)$ is true for at least one $x$ in the domain.
3. $q(x)$ is false for every $x$ in the domain.
4. $q(x)$ is false for at least one $x$ in the domain.

**Example.** Let $\mathcal{P}_1$ denote the set of all polynomials of degree one

$$\mathcal{P}_1 = \{f \mid f(x) = ax + b, \quad a \neq 0\}$$

For $f \in \mathcal{P}_1$, let $p(f)$ be the condition

$$f(x) \text{ crosses the } x\text{-axis}$$

On the other hand, for $f \in \mathcal{P}_1$, the condition

$$q : f'(x) > 0$$

**Definition 1.15.** *Let* $p(x)$ be a condition over a domain $D$.

The statement "$p(x)$ is true for every $x$ in the domain $D$" is denoted as

$$(\forall x \in D)\ p(x).$$

We refer to the symbol $\forall$ as the <u>universal quantifier</u>, and we say $(\forall x \in D)\ p(x)$ is a <u>universally quantified statement</u>.

The statement "$p(x)$ is true for at least one $x$ in the domain $D$" is denoted as

$$(\exists x \in D)\ p(x).$$

We refer to the symbol $\exists$ as the <u>existential quantifier</u>, and we say $(\exists x \in D)\ p(x)$ is an <u>existential quantified statement</u>.

## Negation of quantifiers

$$\sim [(\exists x \in D)\ \ p(x)] \quad \equiv \quad (\forall x \in D)\ \ \sim p(x)$$

Why?

Similarly

$$\sim [(\forall x \in D)\ \ p(x)] \quad \equiv \quad (\exists x \in D)\ \ \sim p(x)$$

**Example 1.16.** Express the following statement in the language of formal logic.

There exists a natural number $n$ so that $2n$ is odd.

**Example.**

Every vector space of dimension 3 has a basis.

**Example.**

There exists a vector space of dimension 3 with no basis.

**Example 1.18.**

For every $x \neq 0$ and every $y \neq 0$, the product of $x$ and $y$ is not zero.

**Example.** Let $p(f, g)$ be the condition "$g(x)$ is the first derivative of $f(x)$"

For instance, $p(x^2, 2x)$ is

For instance, $p(x^3 + 1, 2x)$ is

Fix $f(x) = x^2 + 1$. Then

We know from Calculus that for every polynomial $f(x)$, its derivative is another polynomial. Hence

**Example.** Consider the condition $q : x + 1 > y$ and the statement

$$(\forall x \in \mathbb{R})[(\exists y \in \mathbb{R})\ q(x, y)]$$

Now consider the statement

$$(\exists x \in \mathbb{R})[(\forall y \in \mathbb{R})\ q(x, y)]$$

**Aside.** Which of the following is easier for you to understand?

- $\because (\exists A^{-1} \Leftrightarrow \det A \neq 0) \wedge \det A = 0 \quad \therefore \sim (\exists A^{-1})$
- Recall that $A^{-1}$ exists if and only if $\det A \neq 0$. Since $\det A = 0$, it then follows that $A^{-1}$ does not exist.

# Some definitions from number theory

**Definition.** Let $n$ be an integer.

- We say $n$ is <u>odd</u> if $(\exists k \in \mathbb{Z})\ \ n = 2k + 1$.
- We say $n$ is <u>even</u> if $(\exists k \in \mathbb{Z})\ \ n = 2k$.

**Definition.** Let $n$ be an integer and let $d$ be a positive integer.
We say $d$ <u>divides</u> $n$, and write $d|n$, if $(\exists k \in \mathbb{Z})\ \ n = dk$.

**Definition.** Let $p > 1$ be an integer. We say $p$ is <u>prime</u> if the only positive integers $d$ which divides $p$ are $p = 1$ and $p = d$.

**Definition.** Let $n$ be an integer. Let $d$ and $r$ be integers with $d > 0$ and $0 \leq r < d$.
We say <u>$n$ divided by $d$ gives a remainder $r$</u> if

$$(\exists k \in \mathbb{Z})\ \ n = dk + r.$$

**Definition.** Let $x$ be a real number. We say $x$ is <u>rational</u> if there exist integers $p$ and $q \neq 0$ such that $x = p/q$. If $x$ is not rational, then we say that $x$ is <u>irrational</u>.

## 1.3 Mathematical Proofs

Most mathematical statements (i.e. theorems) that we encounter appear to be of the form

$$\text{If } p, \text{ then } q.$$

For example:

**Theorem.** *Let $f: \mathbb{R} \to \mathbb{R}$. If $f$ is differentiable, then $f$ is continuous.*

**Theorem.** *Let $f: \mathbb{R} \to \mathbb{R}$. If $f$ is a quadratic function, then $f$ crosses the $x$-axis at most twice.*

**Theorem.** *Let $V$ be a vector space. If $V$ is finite-dimensional, then $V$ has a basis.*

These statements are in fact conditional statements depending on a variable:

$$(\forall f \in \mathcal{F})\ \ p(f) \Rightarrow q(f)$$

**We want to show the following:**

**Theorem.** $p \Rightarrow q$

Let's look at the truth table for $p \Rightarrow q$:

**Definition 1.20.** *A <u>mathematical proof</u> is a sequence of statements where each statement is:*

- *a declaration of notation,*
- *a premise, or*
- *a true statement that follows from:*
  - *a definition,*
  - *algebra, or*
  - *previous true statements.*

**Definition 1.21.** *A <u>formal mathematical proof</u> is a finite sequence of statements*

$$A_1, A_2, A_3, \dots, A_n$$

*such that each $A_i$ is either:*

- *known or assumed true or*
- *an be inferred from a known or assumed true statement $A_j$ with $j < i$ (i.e. a previous statement).*

A **good practical definition** of a mathematical proof is:

*An argument in favor of a mathematical statement that will convince a preponderance of knowledgeable peers of its truth.*

**Why do we care about proofs?**

- To communicate mathematics to our peers (and to ourselves).
- To verify the truth of mathematical claims and avoid errors.

**Theorem 1.19.** *Let $x$ be an integer. If $x$ is even, then $x^2$ is even.*

The hypothesis $p(x)$ is:

The conclusion $q(x)$ is:

Scrap work:

**Theorem 1.19.** *Let $x$ be an integer. If $x$ is even, then $x^2$ is even.*

*Proof.* (formal)

1.

2.

3.

4.

5.

6.

7.

**Theorem 1.19.** *Let $x$ be an integer. If $x$ is even, then $x^2$ is even.*

*Proof.* (semi-formal)

1.

2.

3.

4.

5.

6.

7.

**Theorem 1.19.** *Let $x$ be an integer. If $x$ is even, then $x^2$ is even.*

*Proof.* (informal)

Let $x$ be an even integer. Since $x$ is even there exists an integer $k$ such that

$$x = 2k.$$

Computing, we find

$$x^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

Thus there exists an integer $\ell$ such that $x^2 = 2\ell$; in particular, $\ell = 2k^2$. Therefore, $x^2$ is even. Therefore, if $x$ is even, then $x^2$ is even.

**Theorem 1.22.** *Let $x$ and $y$ be integers. If $x$ and $y$ are even, then $(x+y)^2$ is even.*

The hypothesis is:

The conclusion is:

Scrap work:

**Theorem 1.22.** *Let $x$ and $y$ be integers. If $x$ and $y$ are even, then $(x+y)^2$ is even.*

*Proof.* (formal)

1.

2.

3.

4.

5.

6.

7.

8.

9.

10.

11.

12.

13.

**Theorem 1.24.** *Let* $x \in \mathbb{Z}$. *If* $x^2 - 6x + 5$ *is even, then* $x$ *is odd.*

*Attempted proof.* (formal)

1.

2.

3.

4.

5.

6.

7.

**Proof by Contraposition.** Recall that

$$p \Rightarrow q \quad \equiv \quad \sim q \Rightarrow \sim p$$

**Theorem 1.24.** *Let* $x \in \mathbb{Z}$. *If* $x^2 - 6x + 5$ *is even, then* $x$ *is odd.*

The contrapositive is:

Scrap work:

**Theorem 1.24.** *Let* $x \in \mathbb{Z}$. *If* $x^2 - 6x + 5$ *is even, then* $x$ *is odd.*

*Proof.* We proceed by contraposition.

1.

2.

3.

4.

5.

6.

7.

Consider the following sequence of statements:

| | |
|---|---|
| 1. $\frac{1}{2}$ is an integer. | (premise) |
| 2. $1 + \frac{1}{2}$ is an integer. | (1, integers are closed under addition) |
| 3. $\frac{3}{2}$ is an integer. | (2) |
| 4. There is an integer between 1 and 2. | (3) |

**Proof by contradiction.**

**Theorem 1.25.** *Let* $a$ *and* $b$ *be real numbers. If* $a$ *is a rational number and* $b$ *is an irrational number, then the sum* $a + b$ *is irrational.*

To summarize what we have discussed, let's write out what generic direct proofs, proof by contraposition, and proofs by contradiction look like.

**Theorem.** $(\forall x \in D)\ \ p(x) \Rightarrow q(x)$

*Direct proof.*

1. Let $x \in D$ such that $p(x)$ is true. (premise)

2.

3.

4.

5. $q(x)$

6. $(\forall x \in D)\ \ p(x) \Rightarrow q(x)$ (1., 5.)

**Theorem.** $(\forall x \in D)\ \ p(x) \Rightarrow q(x)$

*Proof by contraposition.*

1. Let $x \in D$ such that $\sim q(x)$ is true. (premise)

2.

3.

4.

5. $\sim p(x)$

6. $(\forall x \in D)\ \ \sim q(x) \Rightarrow \sim p(x)$ (1., 5.)

7. $(\forall x \in D)\ \ p(x) \Rightarrow q(x)$ (contrapositive of 6.)

**Theorem.** $p$

*Proof by contradiction*.

| | | |
|---|---|---|
| 1. | $\sim p$ | (premise) |
| 2. | | |
| 3. | $r$ | |
| 4. | | |
| 5. | | |
| 6. | $\sim r$ | |
| 7. | $r \wedge (\sim r)$ | (3., 6.) |

We can write proofs involving multiple cases using mini-direct proofs.

**Theorem.** *If $a$ divided by* 8 *gives a remainder of either* 1 *or* 7*, then $a^2$ divided by* 8 *gives a remainder of* 1.

**Proof.** Let $a$ be an integer such that $a$ divided by 8 gives a remainder of 1 or 7. We shall consider the cases where the remainder is 1 or is 7 separately.

*Case 1: $a$ divided by 8 gives a remainder of 1.*

| | | |
|---|---|---|
| 1. | $(\exists n \in \mathbb{Z})\ a = 8n + 1$ | (premise) |
| 2. | ??? | (algebra) |
| 3. | $(\exists m \in \mathbb{Z})\ a^2 = 8m + 1$ | (3) |

*Case 2: $a$ divided by 8 gives a remainder of 7.*

| | | |
|---|---|---|
| 1. | $(\exists n \in \mathbb{Z})\ a = 8n + 7$ | (premise) |
| 2. | ??? | (justification) |
| 3. | $(\exists m \in \mathbb{Z})\ a^2 = 8m + 1$ | (3) |

*Conclusion:* Therefore, if $a$ divided by 8 gives a remainder of either 1 or 7, then $a^2$ divided by 8 gives a remainder of 1.

**Examples.** The statement

$$(\exists x \in D)\, p(x)$$

is true precisely when $p(x)$ is true for at least one member of $D$.

Hence we can prove the statement by finding one $a$ for which $p(a)$ is true.
We call such a value $a$ an <u>example</u>.

**Example.**

There exists a linear function which passes through the origin.

**Counter-examples.** The statement

$$(\forall x \in D)\, p(x)$$

is true precisely when $p(x)$ is true for every member of $D$.

Hence we can disprove the statement by finding one $a$ for which $p(a)$ is false. We call such a value $a$ a <u>counter-example</u>.

**Example.**

Every prime number is odd.

Consider the following sum:

$$s(n) = \sum_{i=0}^{n} 2^i = 2^0 + 2^1 + 2^2 + \cdots + 2^n$$

When $n$ is small, it is easy to compute the values of $\mathrm{s}(n)$.

**The Principle of Mathematical Induction.**

*Let* $p(n)$ *be a condition over* $\mathbb{N}$. *If the following two statements are true, then* $p(n)$ *is true for each integer* $n \in \mathbb{N}$.

1. (Base case) $p(0)$ *is true;*
2. (Inductive step) *For each* $k \in \mathbb{N}$, *if* $p(k)$ *is true, then* $p(k+1)$ *is true.*

**Theorem 1.27.** *For every* $n \in \mathbb{N}$

$$\sum_{i=0}^{n} 2^i = 2^{n+1} - 1$$

*Proof.*

**The Principle of Strong Mathematical Induction.**

*Let* $p(n)$ *be a condition over* $\mathbb{N}$. *If the following two statements are true, then* $p(n)$ *is true for each integer* $n \in \mathbb{N}$.

1. (Base case) $p(0)$ *is true;*
2. (Inductive step) *For each* $k \in \mathbb{N}$, *if each of* $p(0), p(1), \ldots p(k)$ *is true, then* $p(k+1)$ *is true.*

Recall that a prime number is an integer $n \geq 2$ such that the only positive divisors of $n$ are 1 and itself.

Any integer $n \geq 2$ that is not prime is called composite. For any composite number $n$ there exists integers $2 \leq a, b < n$ such that $\mathrm{n} = \mathrm{a}b$.

**Theorem 1.28.** *Every integer* $n \geq 2$ *is prime or can be expressed as a product of prime numbers.*

*Proof.*

Note that induction can start at any base case.

Suppose we want to prove a condition $p(n)$ holds true for all integers $n \geq 2$.

Then applying the **Principle of Strong Mathematical Induction** to prove the condition $q(n)$ holds true for all integers $n \geq 0$ gives us:

*If the following two statements are true, then* $p(n)$ *is true for each integer* $n \geq 2$.

1. (Base case) $p(2)$ *is true;*
2. (Inductive step) *For each integer* $k \geq 2$, *if each of* $p(2), p(3), \ldots p(k)$ *is true, then* $p(k+1)$ *is true.*

**Theorem 1.29.** *Every integer* $n \geq 2$ *is prime or can be uniquely expressed as a product of prime numbers.*